INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

CHARGERS, BATTERY, LIFT TRUCK AND PALLET TRANSPORTER BATTERY SERVICE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers solid-state battery chargers for charging valve regulated lead-acid (VRLA) and flooded lead-acid (FLA) batteries used for the propulsion and operation of industrial type lift trucks and pallet transporters aboard U.S. Navy ships and at shore stations.

2. CLASSIFICATION. Battery chargers shall be classified as specified in 2.1.

2.1 Type.

a. Type 1 – Single unit for charging 12-cell, 18-cell, or more (24, 36, and 48 volts nominal minimum range); 600- to 1200-ampere-hour capacity batteries with FLA, VRLA, absorbent glass mat (AGM), or gelled electrolyte (GEL) chemistries.

b. Type 2 – Dual unit for charging 12-cell, 18-cell, or more (24, 36, and 48 volts nominal minimum range); 600- to 1200-ampere-hour capacity batteries with FLA, VRLA, AGM, or GEL chemistries.

3. SALIENT CHARACTERISTICS.

3.1 Commercial standard. The charger shall meet UL 1564 except as specified herein.

3.2 Direct current (DC) output characteristics. The output voltage and current of the charger shall be within the minimum ranges required for batteries listed for type 1 and 2 chargers (see 2.1). Additional capability at higher voltages is acceptable, but not required, if standard on the commercial-off-the-shelf unit.

3.3 Operating and display features. The required minimum operating and display features shall be as follows:

a. Message display with a minimum of 16 characters for the display of voltage, current, temperature, and other parameters of interest to the user, including battery identification, history, charge profile, and data provided by the charger or monitoring system module as specified herein.

b. Operate with the charger manufacturer's battery identification device (BID) system (see 3.3.3) to provide information such as battery type identification, serial number, battery parameters, charge profile, and charge and discharge history, if available.

c. Automatic recognition of battery nominal voltage (24, 36, or 48 volts) and selection of charging voltage prior to the start of charging.

d. Automatic temperature recognition and compensation of charging voltage based on actual battery temperature during charging.

e. Automatic alarm and charge start disabling when the battery temperature is too high for normal charging.

f. Manual override of alarm and charge disabling features.

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g. When actual battery temperature is not available, manual entry of a starting battery temperature shall be accommodated.

h. When no charging profile is selected, manual entry of output voltage and current limit steps in a constant current (I), constant voltage (E), and constant current (I) (I-E-I) shall be accommodated.

3.3.1 Additional features. Commercial features not required herein, such as those listed below, may be provided as part of a standard design. These features shall not interfere with required operational characteristics specified herein.

a. Input power failure recovery, which allows the battery to return to the correct charge state after electrical power has been restored, instead of restarting the charge from the beginning in automatic mode.

b. Selectable time delay feature for cool-down of a hot battery.

c. A computer interface connection to a printer or laptop computer.

d. Selectable fast charge feature for charging FLA, VRLA, AGM, and GEL batteries when allowed by the battery manufacturer.

3.3.2 Alternating current (AC) input characteristics. The charger shall operate from a type I, 450-volt AC, 50- or 60-hertz, three-phase delta, ungrounded power source in accordance with MIL-STD-1399-300-1.

3.3.3 BID and monitoring system. The BID system shall be reprogrammable. The system shall work with the battery charger and digital interface devices to provide tabulation of battery charge and discharge history either maintained in the BID installed on the battery or in the charger for download to a laptop computer or similar device. The number of required BID modules shall be as specified (see 7.2). The BID system shall have the minimum characteristics as follows:

a. Battery identification (VRLA, AGM, GEL, or FLA type), manufacturer, and serial number or other identification.

b. Battery voltage and temperature provided to the charger.

c. If battery current is a BID data feature, the shunt shall be bolted in line with the positive or negative lead in VRLA and FLA batteries.

d. BID module data interface between the charger and laptop computer shall be universal serial bus (USB) port connection, RS-232 connection, or equal. BID module data transfer to a handheld digital device shall be by USB port connection.

3.4 Interface requirements.

3.4.1 Dimensions. Battery chargers shall not exceed the following dimensions:

a. Width: 29.5 inches

b. Height: 38.0 inches

c. Depth: 26.0 inches

d. Mass: 400 pounds for a type 1 charger and 900 pounds for a type 2 charger

3.4.2 Input phase balance and grounding. The input supply current shall be balanced to the extent that the current in any phase shall not differ by more than 10 percent from the average current in the three phases. The battery charger shall not establish an electrical ground or require grounding of the power source, either directly or by means of filter capacitors, in accordance with the system grounding requirements of MIL-STD-1399-300-1 for user equipment.

3.4.3 Charger output cabling. Each charger shall be fitted with a cable and connector assembly to connect the battery charger output terminals to the battery connector.

3.4.3.1 Cable. Unless otherwise specified (see 7.2), the cable length shall be 14 or 30 feet and meet the following requirements:

a. Length: 14 feet. The battery charger output cable shall conform to UL 1581, type W, two-conductor, AWG 2/0 portable power cable. The cable shall meet the interior U.S. Navy ship low smoke, halogen-free fire safety test requirements of MIL-DTL-24643. Unless otherwise specified (see 7.2), the 14-foot cable shall contain a minimum of two conductors and may be multi-stranded to improve flexibility.

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b. Length: 30 feet. The battery charger output cable shall conform to UL 1426, BC-5W2, two-conductor, multi-stranded conductor, AWG 3/0 portable marine grade battery power cable. The cable shall meet the interior U.S. Navy ship low smoke, halogen-free fire safety test requirements of MIL-DTL-24643. Unless otherwise specified (see 7.2), the 30-foot cable shall contain a minimum of two conductors and shall be multi-stranded to ensure flexibility.

3.4.3.2 Connectors. The output connector for the battery chargers to the battery shall be Anderson Power SBX 350, or equal, in the color gray (ashore only), or Euro A32 Series, or equal, connectors with auxiliary contacts for BID module connection (afloat only) and shall be provided as specified (see 7.2). (Anderson Power [CAGE 034K5] can be reached by phone at 978-422-3800 or online at www.andersonpower.com.) Adapters needed to connect to batteries with connectors of a different color and physical shape, usually associated with dual voltage chargers, shall be supplied when specified (see 7.2).

3.5 Environmental requirements.

3.5.1 Shock. The battery charger shall be grade B shock qualified for light or medium weight classification as required by the weight of the charger in accordance with MIL-DTL-901. Grade A shock qualification in accordance with MIL-DTL-901 is desired, but not required.

3.5.2 Ambient temperature. The battery chargers shall operate in an ambient temperature range of 32 to 120 °F.

3.5.3 Electromagnetic interference (EMI). The charger and the BID systems shall meet the following requirements of MIL-STD-461: CE101, CE102, CS101, CS114, CS116, RE101, RE102, and RS103. The acceptable minimum is to meet the requirements of MIL-STD-461 through the use of the supplier's input filter. To meet the requirements of MIL-STD-461, the charger and the BID system shall be tested together under a load. To meet the requirements of RE102, the supplier may need to provide additional shielding behind the display and tactile switch button modules to prevent radiated emissions from passing through the display device. This shielding may take the form of a sheet metal enclosure, foil, screening, spray-on conductive metallic coating, or other means. The use of shielded output cables is permitted.

3.5.4 Corrosion resistance. All metal parts of the battery chargers shall be fabricated from corrosion-resistant materials or have an acceptable corrosion-resistant treatment (see 7.2). Circuit cards and other internal components shall be protected from a salt-air environment.

3.5.5 Ventilation. Ventilation of the battery chargers shall be natural draft or with fan assist. Cooling shall provide adequate flow such that under normal operating conditions, the maximum temperature rating of any component is not exceeded. Cooling fans shall have a continuous duty cycle. Filters are required for forced ventilation and shall be reusable and washable.

3.5.6 High-temperature protection. Protection shall be provided to avoid component damage from high temperatures that may arise upon failure of ventilation or from operation in a high-temperature environment. Such protection shall automatically de-energize the battery charger at the DC output. The AC input shall remain energized to operate the charger and its cooling fans.

3.6 Operational requirements.

3.6.1 Current adjustment. The DC output current of the battery chargers shall be capable of automatic and manual adjustment to provide different maximum output currents for different ampere-hour capacity batteries.

3.6.2 Operation (battery charging). Battery chargers shall automatically regulate the output current and voltage for the type of battery being charged as recognized by the BID system or as set into the charger manually by the operator. Normal charges shall be completed in 8 to 8.5 hours from 80 percent depth of discharge, and equalizing charges shall be completed in an additional 3 hours using electronic timers or dV/dt technology (see 7.4.1) to terminate the equalizing charge. The charger shall automatically adjust the charging voltage limit for battery temperature to prevent excessive gassing or undercharging of the battery.

3.6.2.1 Fast charging. When provided, the fast-charging feature (see 3.3.1) shall be capable of charging a discharged battery in 4 hours or less. The fast-charging feature shall not be automatic and shall be manually selectable in order to maintain guarantees by battery suppliers to the U.S. Navy who prohibit the use of fast charging for VRLA batteries.

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3.6.3 Temperature compensation. Charging voltage shall be automatically compensated for variations in actual battery temperature by adjusting the voltage -3 to -4 millivolts per cell for each 1 °F differential in temperature from 80 °F.

3.6.4 Mis-operation. Neither the battery charger nor the operator shall sustain damage or injury when the charger is:

a. Connected with incorrect polarity to any battery.

b. Connected to any battery, regardless of the setting of the control switches, and regardless of whether the AC input is energized.

c. Disconnected from any battery while charging is in progress.

d. Energized without having the output cable connected to a battery.

3.6.5 Remote sensing. Remote sensing of battery voltage and battery temperature shall be provided for battery charger operation to compensate for cable voltage drop and to provide accurate temperature compensation for the charging voltage.

3.6.6 Power switch. The battery chargers shall be provided with a power switch to provide for on-off operation of the charger at the AC input circuit.

3.6.6.1 Power-on indication or light. Battery chargers shall be provided with a light or indication to show when the battery charger is energized.

3.6.7 Indicating devices. The battery chargers shall be furnished with indicating devices as an integral part of the equipment. The devices shall provide, as a minimum, an indication of output current and output voltage. These indications may be integrated into the display system required in 3.3.

3.6.8 Overload protection. Overload protection shall be provided in the DC load circuit. The overload protection shall also serve to protect battery charger components in the event of reversed polarity battery connections.

3.6.9 Mounting. Battery chargers shall be deck or bulkhead mounted. The charger enclosure shall be grounded or bonded for personnel safety in accordance with MIL-STD-1310.

3.6.10 Lifting during installation. Battery chargers shall be capable of being lifted while being handled during installation. Straps or attachment points for lifting shall be located such that when lifting vertically, the battery charger does not tilt more than 5 degrees from the vertical.

3.7 Prohibited components. The following components shall not be used:

a. Variable transformers (typically Variacs)

b. Meter relays

c. Electron tubes or other thermionic devices

d. Auto-transformers or open-delta connected transformers

3.8 Maintainability. Maintenance and repair of battery chargers shall be able to be accomplished within 1 hour.

3.9 Identification plates. The battery chargers shall be provided with a non-corrosive, permanently installed identification plate providing the following information:

a. Nomenclature

b. Type 1 or type 2

c. National stock number

d. Component identification number

e. Contract or order number

f. Electrical characteristics

g. Manufacturer’s serial number

h. Manufacturer’s name and address

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3.9.1 Warning plate. The battery charger shall be provided with a "WARNING" plate that alerts personnel of any potential danger of the battery charger and specifies the safety precautions that shall be observed while operating it. The specifics of potential hazards shall be identified along with directions on what to do if a mishap occurs.

3.10 Prohibited materials. Components shall be free of asbestos, mercury and mercury compounds, polyvinyl chloride, cadmium, and any ozone depleting chemicals.

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. However, used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the battery charger.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Conformance testing. When specified (see 7.2), the battery charger submitted by the manufacturer as meeting the salient characteristics of this CID shall be tested for conformance with the shock requirements of 3.5.1 and the EMI requirements of 3.5.3 by either the Government or a qualified commercial laboratory. This testing is required to determine the need for additional measures to reduce emissions or susceptibility to EMI.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES.

7.1 Source of documents.

7.1.1 Defense specifications and standards. Defense specifications and standards are available online at https://quicksearch.dla.mil.

7.1.2 Underwriters Laboratories, Inc. (UL) standards. UL standards are available online at www.ul.com.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Number and type of units required.

c. Number of BID system modules required (see 3.3.3).

d. Length of the charging cable if other than 14 or 30 feet (see 3.4.3.1).

e. The number of conductors in the cable if other than as specified (see 3.4.3.1).

f. Whether output connectors with auxiliary contacts for BID module connection or direct sensor information are required (see 3.4.3.2).

g. Type of battery cable connector adapter, if required (see 3.4.3.2).

h. Acceptable corrosion-resistant treatment, if required (see 3.5.4)

i. Whether shock testing and EMI and input power waveform testing is required (see 5.2).

j. The required location of shock testing and EMI and input power waveform testing (see 5.2).

k. Packaging requirements (see 6).

7.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

7.4 Definitions.

7.4.1 dV/dt technology. The expression dV/dt represents a change in voltage over a change in time. dV/dt technology offers a way to manage voltage spikes in electrical systems by recording the rate of voltage change over time.

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7.5 Key words.

Electrolyte, gelled
Lead-acid, regulated, valve
VRLA

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – SH
Air Force – 71

Preparing activity:
Navy – SH
(Project 6130-2025-002)

Review activities:
Navy – AS, EC
Air Force – 03, 19
DLA – GS

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